LAB REPORT: DATABASE DESIGN (EER MODELING)

CSBB 204: Database Management Systems — Week 9

Name: Aryan Nautiyal **Roll Number:** 251210015

Aim

To design an Enhanced Entity-Relationship (EER) diagram for a Grade Book database that models students, instructors, courses, grading components, grade cutoffs, enrollments, and student scores — capturing generalization/specialization hierarchies, weak entities, and ternary relationships.

Theory

EER Concept	Description
Supertype / Subtype	A generalization hierarchy where a supertype entity (e.g., Person) shares common attributes with its subtypes (e.g., Student, Instructor). Each subtype inherits the supertype's primary key.
Disjoint, Total	A specialization constraint: {d, t} means every supertype instance belongs to exactly one subtype (disjoint) and must belong to some subtype (total).
Weak Entity	An entity that cannot be uniquely identified by its own attributes alone. It depends on an identifying (owner) entity and uses a partial key (discriminator) combined with the owner's primary key.
Identifying Relationship	The relationship between a weak entity and its owner. Represented with a double diamond in Chen notation.
Ternary Relationship	A relationship involving three entity types simultaneously, modeled as an associative entity.
Composite Key	A primary key composed of two or more attributes (e.g., course_number + section_number + term).

Data Requirements Summary

The Grade Book database captures the following:

1. **Students** are identified by a unique student ID, first name, last name, and email.
2. **Instructors** teach courses each term and define grade cutoffs (A/B/C/D/F) per course.
3. **Courses** are uniquely identified by course number, section number, and term.
4. Each course has multiple **grading components** (mid-term, final, project, etc.) with maximum points and percentage weights summing to 100.
5. **Scores** record the points each student earns in each grading component of each course.

Entity & Attribute Description

Entity	Type	Attributes
Person	Supertype	person_id (PK), first_name, last_name, email (UK)
Student	Subtype of Person	person_id (PK, FK → Person)
Instructor	Subtype of Person	person_id (PK, FK → Person), department
Course_Section	Strong Entity	course_number (PK), section_number (PK), term (PK), instructor_id (FK)
Grading_Component	Weak Entity	component_name (Partial Key), max_points, weight; Owner: Course_Section
Grade_Cutoff	Weak Entity	letter_grade (Partial Key), cutoff_score; Owner: Course_Section
Enrollment	Associative (M:N)	student_id (FK), course_number (FK), section_number (FK), term (FK)
Score	Associative (Ternary)	student_id (FK), course_number (FK), section_number (FK), term (FK), component_name (FK), points_earned

Enhanced ER Diagram

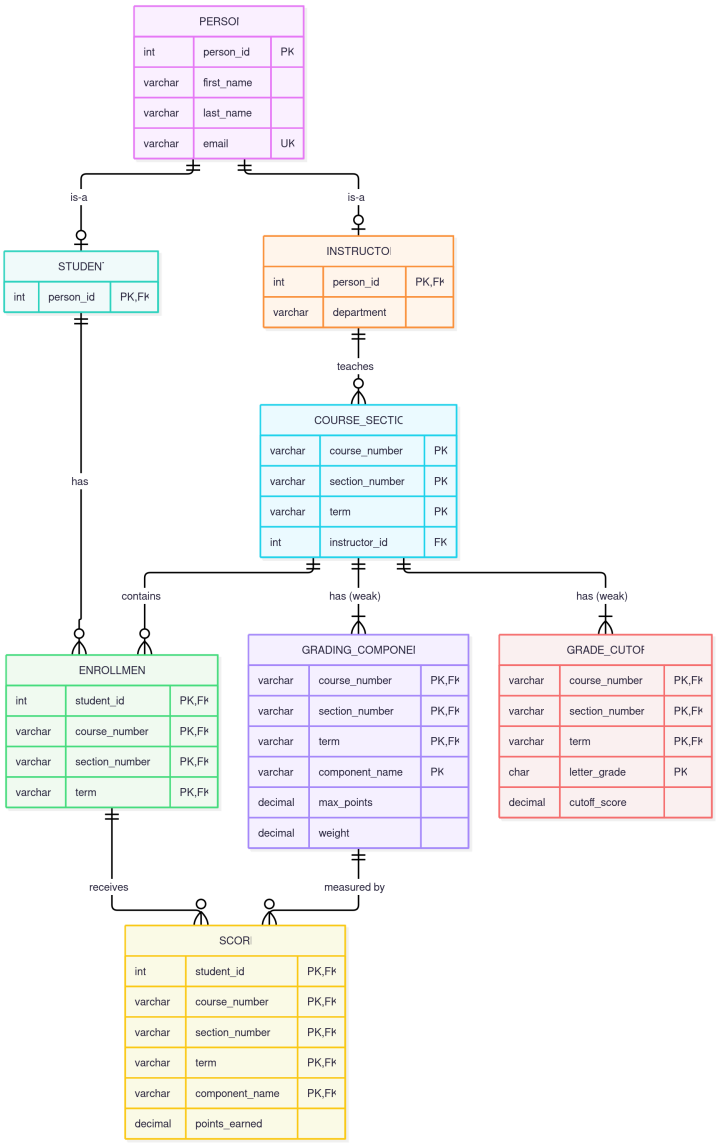


Figure 1: EER Diagram for the Grade Book Database (generated using dbdiagram.io)

Relationships & Cardinality

Relationship	Min:Max (E1)	Min:Max (E2)	Ratio	Notes
Person → Student / Instructor	(1,1)	(0,1)	Super/Sub	{d, t} specialization
Instructor → Course_Section	(1,N)	(1,1)	1 : N	Every section has exactly one instructor
Student → Course_Section	(1,N)	(0,N)	M : N	Via Enrollment entity
Course_Section → Grading_Component	(1,N)	(1,1)	1 : N	Identifying (weak entity)
Course_Section → Grade_Cutoff	(1,N)	(1,1)	1 : N	Identifying (weak entity)
Enrollment → Score	(1,N)	(1,1)	1 : N	Score per component per enrollment
Grading_Component → Score	(1,N)	(1,1)	1 : N	Links component to earned points

Relational Schema Mapping

Primary keys are underlined. Foreign key constraints are listed below each relation.

Person (person_id, first_name, last_name, email)

Student (person_id)
↳ **FK:** `person_id` references **Person**

Instructor (person_id, department)
↳ **FK:** `person_id` references **Person**

Course_Section (course_number, section_number, term, instructor_id)
↳ **FK:** `instructor_id` references **Instructor**

Grading_Component (course_number, section_number, term, component_name, max_points, weight)
↳ **FK:** (`course_number`, `section_number`, `term`) references **Course_Section**

Grade_Cutoff (course_number, section_number, term, letter_grade, cutoff_score)
↳ **FK:** (`course_number`, `section_number`, `term`) references **Course_Section**

Enrollment (student_id, course_number, section_number, term)
↳ **FK:** `student_id` references **Student**
↳ **FK:** (`course_number`, `section_number`, `term`) references **Course_Section**

Score (student_id, course_number, section_number, term, component_name, points_earned)
↳ **FK:** (`student_id`, `course_number`, `section_number`, `term`) references **Enrollment**
↳ **FK:** (`course_number`, `section_number`, `term`, `component_name`) references **Grading_Component**